

The Evolving World Of WIRELESS CHARGING TECHNOLOGY



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Nikola Tesla in the late-18th and early-19th centuries described a device that he believed could transmit electrical power from one conductor to another without the need for wires. But his work in this field failed due to engineering and financial limitations. Soon thereafter power cables became the commonly-accepted means of transporting electricity across distances. But with widespread adoption of small, portable devices with batteries in need of constant recharging, people's attention is again turning to wireless power.

Types of wireless charging

Power through radio waves. Transmitting power through radio waves at first seemed promising. A good example of how radio waves can transmit power is a crystal radio. This basic radio consists of a long wire as

an antenna, a diode, another wire to act as ground wire and a crystal earphone. By attaching the two wires to either end of the diode, connecting the ground wire to a metal stake in the ground and connecting the crystal earphone to the ends of the diode, this radio can pick up radio waves that we can actually hear. We do not need a battery or any other power source to hear these; radio waves themselves provide the power.

The trouble with radio wave transmissions is that these are not very efficient. These spread out as these transmit; only a relatively-small percentage would make it to the antenna for the charging device.

But there are other methods to send electricity wirelessly, including through microwaves or magnetism. Magnet approach really resonated with engineers looking for a way to get rid of those extra charging cords.

Resonant wireless charging. There are mainly two types of wireless charging: inductive and conductive. Inductive charging involves an electromagnetic field to transfer charge across a gap and into the device's battery. Requisite for inductive charging is an induction coil, which is capable of receiving wireless energy. Conductive charging functions through direct electrical contact between the power source and the battery.

Wireless charging or inductive charging uses an electromagnetic field to transfer energy between two objects. This is usually done with a charging station. Energy is sent through an inductive coupling to an electrical device, which can then use that energy to charge batteries or run the device.

Induction chargers typically use an induction coil to create an alternating electromagnetic field within a charging base station. A second induction coil in the portable device takes power from the

